

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Physics

- 1 A magnet rests on a top-pan balance. A straight horizontal wire of length 4.0 cm is held between the magnet's poles and carries current I . The balance reading increases linearly from 500.0 g at $I = 0$ to 504.0 g at $I = 2.0 \text{ A}$. What is the magnitude of the magnetic field strength between the poles? Take $g = 10 \text{ N kg}^{-1}$.
- A 0.0050 T
 B 0.050 T
 C 0.50 T
 D 2.0 T
 E 5.0 T
 F 50 T

- 2 A block of mass 3.0 kg rests on a smooth, horizontal table. It is connected by two light, inextensible strings passing over smooth, frictionless pulleys at opposite ends of the table to two hanging masses of 5.0 kg and 2.0 kg , as shown in the diagram.

The system is released from rest.

What is the tension T_1 in the string attached to the 5.0 kg mass and the tension T_2 in the string attached to the 2.0 kg mass while the blocks are moving?

[Take the acceleration due to gravity to be $g = 10 \text{ m s}^{-2}$.]

	T_1 / N	T_2 / N
A	15	6
B	15	20
C	35	20
D	35	26
E	50	20
F	50	26
G	65	14
H	65	20

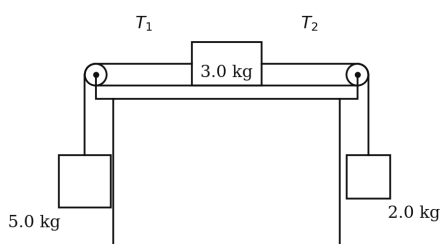


Diagram not to scale

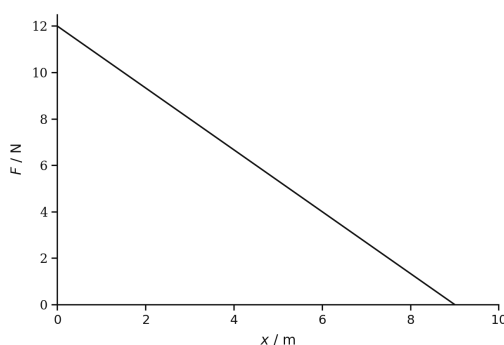
- 3 A laser beam reflects off a plane mirror M_1 and then off a second plane mirror M_2 . The direction of the beam incident on M_1 is fixed. M_1 is rotated by an angle α in a clockwise direction.

To ensure the final beam reflected from M_2 maintains a constant direction, M_2 must be rotated by:

- A α in the clockwise direction
 - B α in the anticlockwise direction
 - C 2α in the clockwise direction
 - D 2α in the anticlockwise direction
 - E $\frac{\alpha}{2}$ in the clockwise direction
- 4 A block of mass 2.0 kg is initially moving at a speed of 4.0 m s^{-1} along a smooth horizontal surface in the positive x -direction.

A single horizontal force F acts on the block in the direction of its motion. The graph shows how F varies with the displacement x of the block.

What is the speed of the block when it reaches $x = 6.0 \text{ m}$?



- A 6.0 m s^{-1}
- B 6.9 m s^{-1}
- C 7.2 m s^{-1}
- D 8.0 m s^{-1}
- E 8.4 m s^{-1}
- F 9.4 m s^{-1}

- 5 The vibration of the $\text{H} - \text{Cl}$ bond can be modelled by assuming the chlorine atom is stationary and the hydrogen atom oscillates with simple harmonic motion.

The frequency of the vibration is 3.0

$\times 10^{13}$

Hz and the amplitude (maximum displacement from the equilibrium position) is 5.0

$\times 10^{-12}$

m .

What is the average speed of the hydrogen atom over one full period of oscillation?

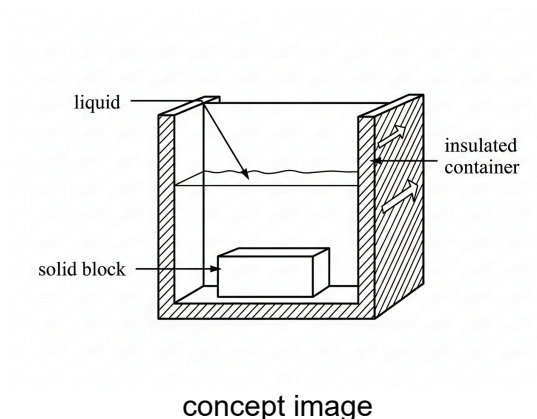
- A 1.5
 $\times 10^2$
 ms^{-1}
- B 3.0
 $\times 10^2$
 ms^{-1}
- C 6.0
 $\times 10^2$
 ms^{-1}
- D 9.4
 $\times 10^2$
 ms^{-1}
- E 3.0
 $\times 10^8$
 ms^{-1}

- 6 A hot solid block is placed into a cold liquid in an insulated container.

The mass of the liquid is twice the mass of the block. The specific heat capacity of the liquid is three times the specific heat capacity of the block.

The temperature of the block decreases by ΔT .

Assuming no thermal energy is transferred to the surroundings, what is the temperature increase of the liquid?



- A $\frac{\Delta T}{6}$
- B $\frac{\Delta T}{5}$
- C $\frac{\Delta T}{3}$
- D $\frac{\Delta T}{2}$
- E $5\Delta T$
- F $6\Delta T$

- 7 The activity of a radioactive source is measured. Its activity after 1 hour is 8 times its activity after 4 hours.

Its activity after 2 hours is 120 Bq.

What was the initial activity of the source in Bq?

- A 30
- B 120
- C 240
- D 480
- E 960
- F 1920

- 8** A sealed syringe with a freely moving, frictionless piston contains a fixed mass of an ideal gas. At a depth of 30 m below the surface of a lake, the volume of the gas in the syringe is 80 cm^3 . When the syringe is moved to a depth of 10 m below the surface, the volume of the gas increases to 120 cm^3 .

The temperature of the water is uniform throughout the lake, and the gas remains in thermal equilibrium with the water.

What is the volume of the gas in the syringe at the surface of the lake?

- A** 135 cm^3
 - B** 140 cm^3
 - C** 150 cm^3
 - D** 160 cm^3
 - E** 180 cm^3
 - F** 200 cm^3
 - G** 240 cm^3
 - H** 360 cm^3
- 9** Two cylindrical rods, X and Y, are used as thermal conductors between two reservoirs maintained at a constant temperature difference. Rod Y has 3 times the thermal conductivity, twice the length, and twice the diameter of rod X.
- In the steady state, the rate of heat flow through rod X is P_X and the rate of heat flow through rod Y is P_Y .

What is the value of the ratio $\frac{P_Y}{P_X}$?

- A** 2
- B** 3
- C** 6
- D** 12
- E** 24

- 10** A block of mass 4.0 kg is placed on a rough plane inclined at an angle of 30° to the horizontal. The coefficient of static friction between the block and the plane is $\mu = \frac{3}{6}$. The block is attached to a light, inextensible string that passes over a smooth, frictionless pulley fixed at the top of the incline and supports a hanging mass M , as shown in the diagram. What is the range of possible values for the hanging mass M such that the system remains in static equilibrium?

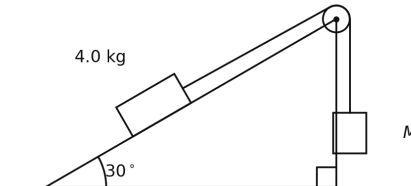


Diagram not to scale

- A** $1.0 \text{ kg} \leq M \leq 3.0 \text{ kg}$
- B** $1.5 \text{ kg} \leq M \leq 2.5 \text{ kg}$
- C** $1.0 \text{ kg} \leq M \leq 2.0 \text{ kg}$
- D** $2.0 \text{ kg} \leq M \leq 3.0 \text{ kg}$
- E** $2.0 \text{ kg} \leq M \leq 4.0 \text{ kg}$

- 11** A U-tube of uniform cross-sectional area, open to the atmosphere at both ends, initially contains a liquid of density ρ_1 .

A second immiscible liquid of density ρ_2 is poured into the left arm, forming a column of height h .

A third immiscible liquid of density ρ_3 is then poured into the right arm until the top liquid surfaces in both arms are at the exact same horizontal level, as shown in the diagram.

Given that $\rho_1 > \rho_2 > \rho_3$, which expression gives the height h_3 of the column of the third liquid?

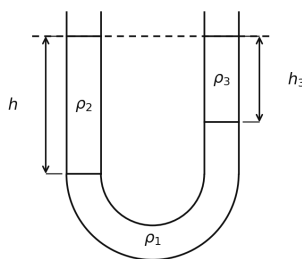


Diagram not to scale

- A** $\left(\frac{\rho_2}{\rho_3}\right) h$
- B** $\left(\frac{\rho_3}{\rho_2}\right) h$
- C** $\left(\frac{\rho_1 - \rho_2}{\rho_1 - \rho_3}\right) h$
- D** $\left(\frac{\rho_1 - \rho_3}{\rho_1 - \rho_2}\right) h$
- E** $\left(\frac{\rho_2 - \rho_3}{\rho_1 - \rho_3}\right) h$
- F** $\left(\frac{\rho_1 - \rho_2}{\rho_2 - \rho_3}\right) h$
- G** $\left(\frac{\rho_1 + \rho_2}{\rho_1 + \rho_3}\right) h$
- H** $\left(\frac{\rho_1 - \rho_2}{\rho_1 + \rho_3}\right) h$
- 12** Plane water waves travel along a straight canal. At a stationary post, wave crests are observed to pass every 2.0 s. A boat travels along the canal in the same direction as the waves with a constant speed of 4.0 m/s. An observer on the boat records that wave crests overtake the boat every 6.0 s.

What is the speed of the waves?

- A** 2.0 m/s
- B** 3.0 m/s
- C** 4.0 m/s
- D** 6.0 m/s
- E** 12.0 m/s

- 13** A small block of mass 2.0 kg is released from rest at the top of a smooth, curved ramp at a height $H = 20 \text{ m}$ above horizontal ground. The end of the ramp is horizontal and is at a height $h = 5.0 \text{ m}$ above the ground, where the block leaves the ramp.

Air resistance is negligible.

The gravitational field strength is $g = 10 \text{ N kg}^{-1}$.

What is the horizontal distance d travelled by the block between leaving the ramp and landing on the ground?

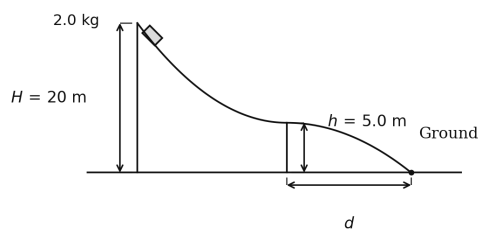


Diagram not to scale

- A** $5\sqrt{3} \text{ m}$
B 10 m
C $10\sqrt{3} \text{ m}$
D 15 m
E 20 m
F $20\sqrt{3} \text{ m}$
- 14** Three identical resistors are connected in a circuit with a DC power supply. One resistor is connected in series with a parallel combination of the other two. What fraction of the total power supplied by the source is dissipated by the parallel combination?
- A** $\frac{1}{9}$
B $\frac{1}{6}$
C $\frac{1}{4}$
D $\frac{1}{3}$
E $\frac{1}{2}$
F $\frac{2}{3}$

- 15** An electric pump lifts oil of density 800 kg m^{-3} through a vertical height of 5.0 m at a constant volumetric flow rate of $2.5 \times 10^{-3} \text{ m}^3 \text{ s}^{-1}$. The pump system has an overall efficiency of 50% and is connected to a 50 V DC power supply.

Assuming $g = 10 \text{ m s}^{-2}$, what is the current drawn from the supply?

- A** 1.0 ext A
- B** 2.0 ext A
- C** 2.5 ext A
- D** 4.0 ext A
- E** 5.0 ext A

- 16** A continuous wave travels from medium P, where its speed is v , into medium Q, where its speed is $0.8v$.

Which of the following gives the fractional change in the wavelength of the wave?

- A** $-\frac{1}{4}$
- B** $-\frac{1}{5}$
- C** 0
- D** $\frac{1}{5}$
- E** $\frac{1}{4}$

- 17** A solid cuboid is made from a material of density 2000 kg m^{-3} . Its side lengths are in the ratio $1 : 2 : 4$. The cuboid is placed on a horizontal surface to exert the maximum possible pressure. This pressure is 8.0 kPa .

What is the mass of the cuboid?

(The gravitational field strength g is 10 N kg^{-1})

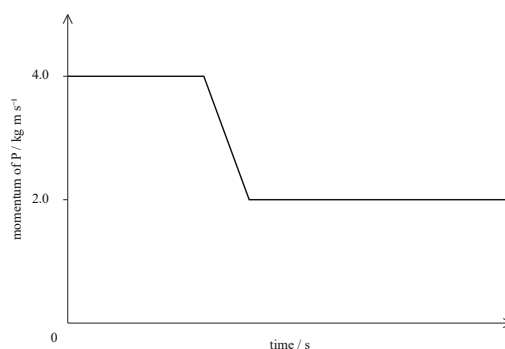
- A** 2.0 kg
- B** 16 kg
- C** 128 kg
- D** 160 kg
- E** 1024 kg
- F** 16000 kg

- 18** A particle is projected from a point O with speed u . The particle passes through a target point P located at a horizontal distance d and vertical height h from O .

Which of the following expressions gives the minimum value of u required for the particle to reach P ?

- A** $\sqrt{g(d+h)}$
B $\sqrt{2g(d+h)}$
C $\sqrt{g(d^2+h^2)}$
D $\sqrt{2g(d^2+h^2)}$
E $\sqrt{g(d^2+h^2)}$
F $\sqrt{g(h^2+d^2+h^2)}$

- 19** An object P collides with a stationary object Q on a straight, frictionless track. The graph shows the momentum of object P as a function of time.



What is the momentum of object Q after the collision?

- A** -4.0 kg m s^{-1}
B -2.0 kg m s^{-1}
C 0 kg m s^{-1}
D 2.0 kg m s^{-1}
E 4.0 kg m s^{-1}
F 6.0 kg m s^{-1}

- 20** An electromagnet is placed on a top-pan balance and produces a uniform horizontal magnetic field of strength 0.50 T .
A horizontal beam of protons is fired through the field, perpendicular to the field lines. The experiment is then repeated with a beam of electrons at the same speed, travelling in the same direction.

At any instant, there are 1.0×10^{10} particles within the magnetic field.

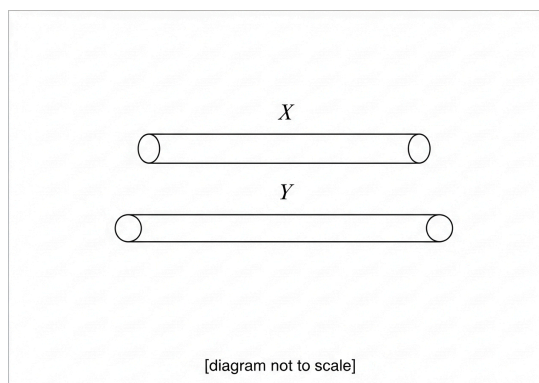
The difference between the two balance readings is 1.6 g .

What is the speed of the particles?

(elementary charge, $e = 1.6 \times 10^{-19} \text{ C}$; gravitational field strength, $g = 10 \text{ N kg}^{-1}$)

- A** $1.0 \times 10^5 \text{ m s}^{-1}$
B $2.0 \times 10^7 \text{ m s}^{-1}$
C $1.0 \times 10^7 \text{ m s}^{-1}$
D $4.0 \times 10^7 \text{ m s}^{-1}$
E $1.0 \times 10^{10} \text{ m s}^{-1}$
- 21** Wire X and wire Y are made of the same material. Wire Y has twice the length and half the diameter of wire X.

What is the ratio of their electrical resistances, $\frac{R_Y}{R_X}$?



concept image

- A** $\frac{1}{8}$
B $\frac{1}{4}$
C $\frac{1}{2}$
D 2
E 4
F 8

- 22** An ideal gas bubble of volume V is released at a depth $3H$ below the surface of a liquid. It rises to a depth H .
The atmospheric pressure at the surface is equal to the pressure exerted by a column of the liquid of height H .
Assuming the temperature of the gas remains constant, what is the volume of the bubble at depth H ?

- A $\frac{V}{3}$
- B $\frac{V}{2}$
- C $2V$
- D $3V$
- E $4V$

- 23** A laser beam from a fixed source is incident on a plane mirror M_1 . The reflected beam then strikes a second plane mirror M_2 . M_1 rotates with angular speed ω and M_2 rotates with angular speed 4ω . Both mirrors rotate in the same direction about parallel axes perpendicular to the plane of incidence.

What is the angular speed of the final beam reflected from M_2 ?

- A 3ω
- B 5ω
- C 6ω
- D 8ω
- E 10ω

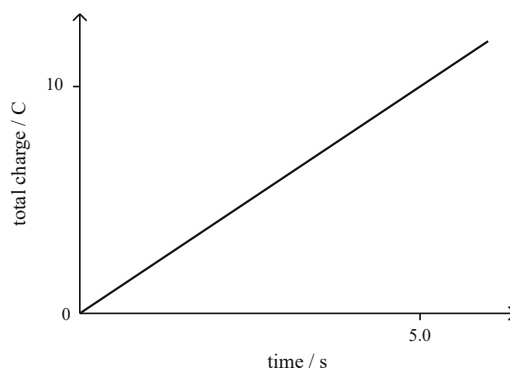
- 24** A solid cuboid of mass 1.0 kg is placed on a horizontal surface. Depending on which face it rests on, it can exert one of three possible pressures: 0.40 N cm^{-2} , 0.50 N cm^{-2} , or 2.0 N cm^{-2} .

What is the density of the cuboid's material, in g cm^{-3} ?

(Take the gravitational field strength g to be 10 N kg^{-1} .)

- A 0.05
- B 0.40
- C 2.0
- D 2.5
- E 20
- F 20000

- 25** A resistor of resistance $4.0\ \Omega$ is connected to a power supply. The graph shows the total charge that has passed through the resistor as a function of time.



What is the potential difference across the resistor?

- A** 0.50 V
 - B** 2.0 V
 - C** 8.0 V
 - D** 20 V
 - E** 25 V
 - F** 40 V
- 26** A charged particle of mass m is released from rest in a uniform field. The particle experiences a constant force F due to the field.

What is the kinetic energy of the particle after it has travelled a distance d parallel to the force?

- A** $\frac{1}{2}Fd$
- B** $2Fd$
- C** Fd
- D** $\frac{Fd}{m}$
- E** Fdm

- 27** The electrical resistance R of a uniform metal wire depends on its length L and cross-sectional area A .

Which of the following proportionalities is correct?

A $R \propto LA$

B $R \propto \frac{A}{L}$

C $R \propto L$

D $R \propto \frac{1}{A}$

E $R \propto \frac{L}{A}$